

Chaotic Rotational Dynamics of Hyperion: Sensitivity, Stability, and Transition to Chaos

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This study investigates the chaotic rotational dynamics of Hyperion, driven by its sensitivity to initial angles (θ), angular frequencies (ω), orbital eccentricity (e), and asphericity (β). Numerical modelling revealed that small changes in initial orientation or angular frequency lead to divergent trajectories, indicative of chaos, while certain configurations exhibit stability. Higher e and β amplify torque variability, destabilising the system, whereas lower values promote regular motion. Poincaré sections and Fourier analysis highlight transitions from periodic to chaotic behaviour, with period-doubling bifurcations providing evidence of non-linear dynamical systems. These findings offer insights into the mechanisms of chaos and stability in celestial bodies, explaining why moons with lower e and β exhibit predictable rotation, while more irregular systems display chaotic motion.

I. INTRODUCTION

Hyperion, a small and irregular moon of Saturn discovered in 1848, exhibits chaotic rotational behaviour that differentiates it from larger, more spherical satellites. While many moons maintain stable spin states due to near-spherical mass distributions and tidal locking, Hyperion's elongated shape and inhomogeneous internal structure result in complex, tumbling motion. Observations from the Voyager and Cassini missions have confirmed that this rotation is highly sensitive to initial conditions, with minor variations in orientation or spin rate producing significantly different long-term outcomes.

This study aims to elucidate the underlying mechanisms responsible for Hyperion's chaotic spin dynamics. It does so by establishing a theoretical framework that couples the moon's orbital motion around Saturn with equations governing its internal rotational state. The model treats the Saturn–Hyperion system as a two-body gravitational problem modified by Hyperion's aspherical mass distribution. By conducting systematic numerical simulations, this approach enables examination of how orbital eccentricity and mass asymmetry affect the transition from regular, quasi-periodic rotation to chaotic, sensitivity-dominated regimes. The analysis thus provides insights into the interplay between orbital parameters, internal structure, and dynamical complexity. These findings may also inform our understanding of other non-spherical satellites and the conditions that lead to chaotic rotational evolution in planetary systems.

II. BACKGROUND

A. Equations of Motion

To investigate the coupled orbital and rotational dynamics of Hyperion, its motion is modelled as a restricted two-body problem. Let M_{Sat} represent Saturn's mass, G

the gravitational constant, and (x_H, y_H) the coordinates of Hyperion in an inertial frame with Saturn fixed at the origin. The equations of motion governing the orbit are:

$$\frac{dx_H}{dt} = v_{xH}, \quad \frac{dy_H}{dt} = v_{yH}, \quad (1)$$

$$\frac{dv_{xH}}{dt} = -\frac{GM_{\text{Sat}}x_H}{(x_H^2 + y_H^2)^{3/2}}, \quad \frac{dv_{yH}}{dt} = -\frac{GM_{\text{Sat}}y_H}{(x_H^2 + y_H^2)^{3/2}}. \quad (2)$$

These equations yield an elliptical orbit when initial conditions are chosen to match a semi-major axis a and orbital eccentricity e .

Hyperion's non-spherical mass distribution is characterised by principal moments of inertia $I_1 < I_2 < I_3$ and the asphericity parameter:

$$\beta = \sqrt{3} \frac{I_2 - I_1}{I_3}. \quad (3)$$

The rotational angle $\theta(t)$ denotes the orientation of the principal axis I_1 . The gravitational torque arising from Saturn's field leads to the following rotational equation:

$$\frac{d^2\theta}{dt^2} = -\frac{GM_{\text{Sat}}\beta^2}{2R_H^3} \sin[2(\theta - \varphi)], \quad (4)$$

where $R_H = \sqrt{x_H^2 + y_H^2}$ is the orbital radius, and φ is the true anomaly. For numerical integration, this equation is rewritten as two coupled first-order differential equations:

$$\frac{d\theta}{dt} = \omega, \quad \frac{d\omega}{dt} = -\frac{GM_{\text{Sat}}\beta^2}{2R_H^3} \sin[2(\theta - \varphi)]. \quad (5)$$

Initial conditions for the orbit are chosen to ensure a stable elliptical trajectory. At periapsis ($t = 0$), the conditions are:

$$x_H(0) = a(1 - e), \quad y_H(0) = 0, \quad (6)$$

$$v_{xH}(0) = 0, \quad v_{yH}(0) = \sqrt{\frac{GM_{\text{Sat}}(1+e)}{a(1-e)}}. \quad (7)$$

Initial rotational conditions $(\theta(0), \omega(0))$ are systematically varied along with β and e to explore the parameter space [1].

The coupled orbital and rotational equations are solved numerically using a suitable integration method, with a sufficiently small time step to ensure accuracy and stability over multiple orbital periods. The orbital solutions are validated against analytical results for elliptical trajectories. The rotational dynamics are then analysed under varying β and e to identify transitions from regular to chaotic behaviour.

III. RESULTS

A. Numerical Modelling of Orbital Dynamics

The governing equations of motion and initial conditions were numerically integrated to model Hyperion's orbit around Saturn. By applying the initial conditions at perihelion and integrating over one orbital period T_H , the simulation produced a stable elliptical trajectory.

Figure 1 shows the normalised x_H - y_H plot over one orbit, confirming that the numerical solution reproduces the expected ellipse for $e = 0.105$. This result validates the initial conditions and numerical approach, as the simulated orbit matches theoretical expectations.

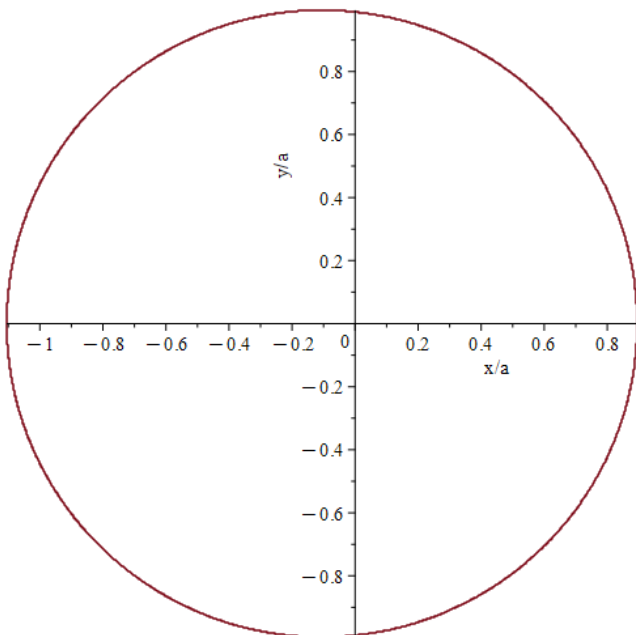


FIG. 1. Normalised orbital trajectory of Hyperion over one orbital period T_H , scaled by the semi-major axis a . The elliptical shape and correct eccentricity confirm the validity of the initial conditions and numerical method.

The long-term stability of the solution was assessed by measuring Hyperion's displacement from its initial position after multiple orbital periods, normalised by a . This displacement indicates the accumulation of numerical errors over time.

Figure 2 illustrates how, with default solver settings, the displacement increases over successive orbits. While acceptable in the short term, this growth highlights the accumulation of errors, a common issue in long-term simulations.

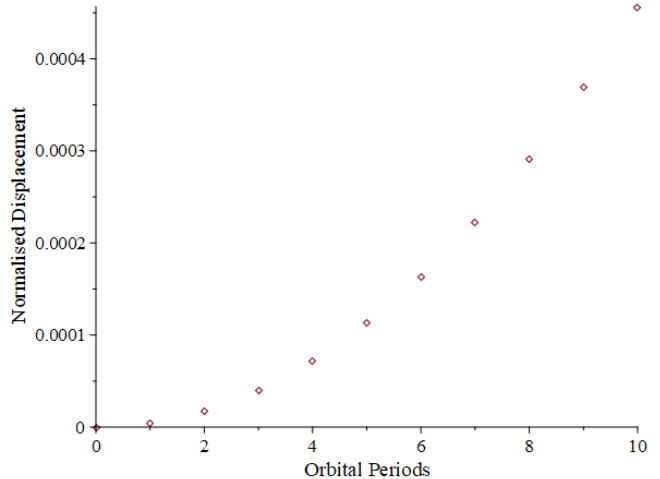


FIG. 2. Normalised displacement of Hyperion from its initial position at perihelion, measured at integer multiples of the orbital period T_H . With default solver settings, the displacement grows over successive orbits, indicating the build-up of numerical errors.

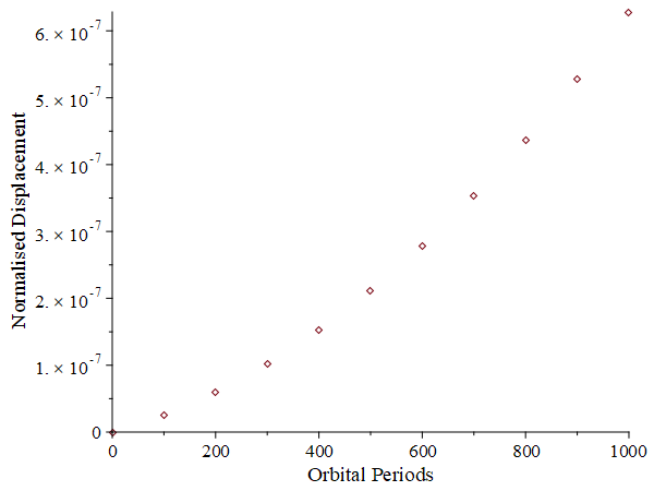


FIG. 3. Normalised displacement of Hyperion over multiple orbital periods, computed with tighter tolerances and a more accurate integration method. Compared to Fig. 2, the growth in displacement is substantially reduced, indicating improved numerical stability and accuracy.

To address this, solver tolerances were refined,

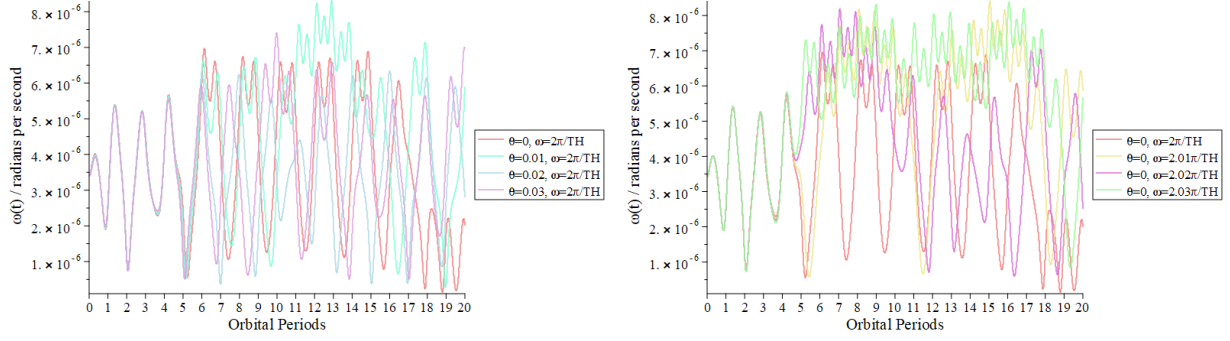


FIG. 4. Comparison of $\omega(t)$ over 20 orbital periods T_H : **(Left)** Evolution of $\omega(t)$ for small variations in $\theta(0)$, with the same initial $\omega(0)$. Diverging trajectories indicate sensitivity to initial orientation, characteristic of chaos. **(Right)** Evolution of $\omega(t)$ for fixed $\theta(0)$ and slightly varied $\omega(0)$. Small changes in $\omega(0)$ result in significant differences, illustrating the system's sensitivity.

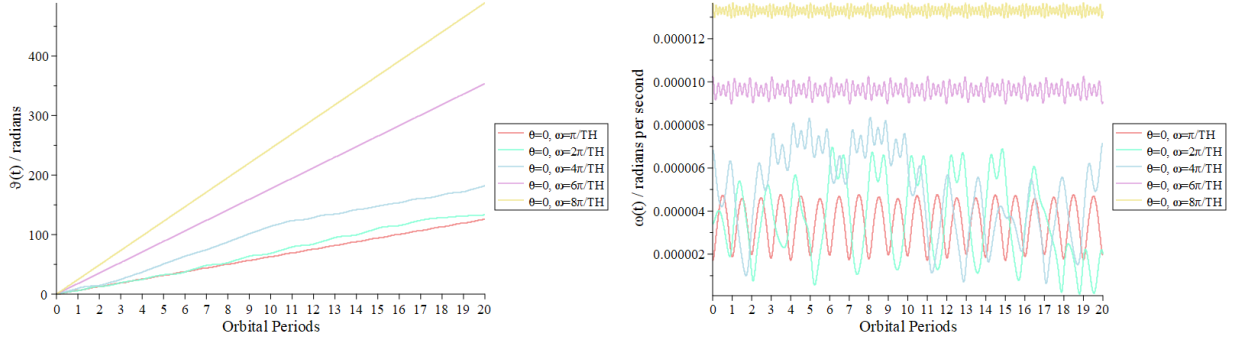


FIG. 5. Comparison of $\theta(t)$ and $\omega(t)$ for varying $\omega(0)$: **(Left)** Evolution of $\theta(t)$ for initial $\omega(0)$ values as multiples of ω_H . Stable values produce smoother growth, while unstable ones deviate significantly. **(Right)** Evolution of $\omega(t)$ for the same initial values. Stable cases show clear stable periodicity, while unstable cases exhibit chaotic fluctuations.

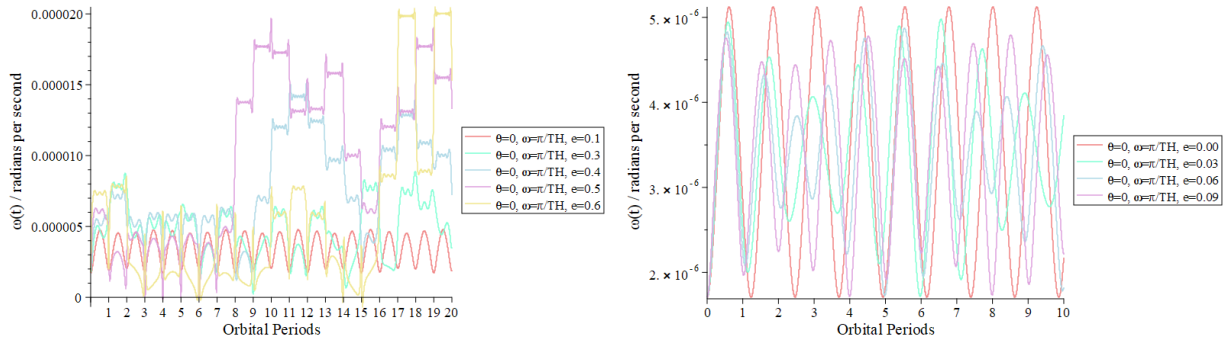


FIG. 6. Effect of orbital eccentricity e on $\omega(t)$: **(Left)** Evolution of $\omega(t)$ for larger e values. Increased variability in torque drives chaotic rotational behaviour. **(Right)** Evolution of $\omega(t)$ for smaller e values. Reduced torque fluctuations allow for more regular, stable dynamics.

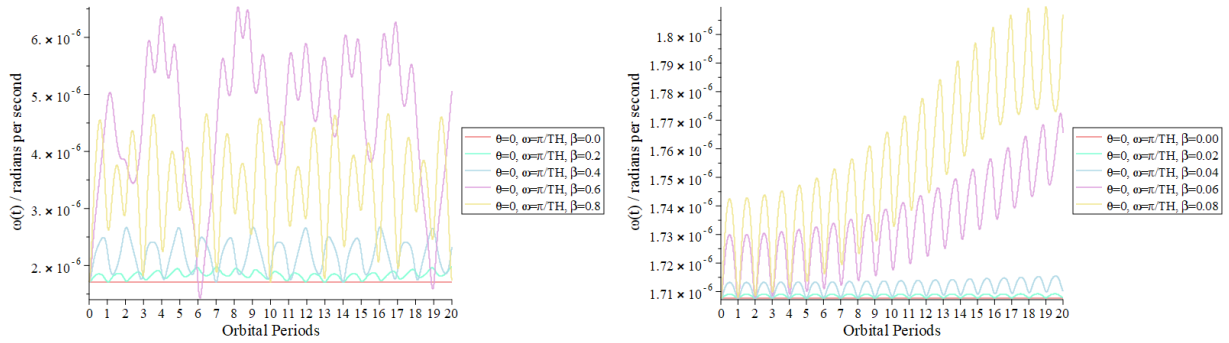


FIG. 7. Impact of asphericity β on $\omega(t)$: **(Left)** Evolution of $\omega(t)$ for larger β . Increased asphericity amplifies chaotic motion. **(Right)** Evolution of $\omega(t)$ for smaller β . A more spherical shape promotes stable dynamics.

and a more accurate numerical method (adaptive Runge–Kutta–Fehlberg) was employed [2]. Figure 3 demonstrates the improved stability, with significantly reduced displacement growth over many more orbits. This highlights the importance of optimising numerical parameters for extended integrations.

With the orbital solution validated, these results were coupled with the rotational dynamics to examine how Hyperion’s asphericity and gravitational torques produce its chaotic spin state.

B. Investigation of Chaotic Dynamics in Rotation

This section explores Hyperion’s rotational dynamics by integrating its equations of motion under various initial conditions and parameters. Figures illustrate how changes in $\theta(0)$, $\omega(0)$, orbital eccentricity e , or asphericity β influence the behaviour of $\theta(t)$ and $\omega(t)$ over multiple orbital periods T_H .

Figures 4 and 5 demonstrate Hyperion’s sensitivity to initial conditions. Small changes in $\theta(0)$ or $\omega(0)$ lead to significant divergence in $\omega(t)$ over time, a hallmark of chaotic dynamics. However, certain initial angular frequencies, especially integer multiples of the angular frequency ω_H , produce smoother growth in $\theta(t)$ and more stable values for $\omega(t)$. This synchronisation between orbital and rotational dynamics reduces chaos, stabilising the motion.

The effect of e is shown in Figure 6. Higher e increases variability in the gravitational torque as the distance to Saturn fluctuates, leading to chaotic tumbling. For lower e , where the orbit is more circular, torque fluctuations are minimised, producing regular and stable motion. These findings align with the expectation that circular orbits reduce perturbations and simplify rotational behaviour.

Figure 7 investigates the impact of β . Larger β , representing greater mass irregularity, results in stronger torques and chaotic motion. As β approaches zero, Hyperion becomes more spherical, and the rotational dynamics stabilise, resembling a tidally locked moon with predictable behaviour.

These results show that Hyperion’s stability depends on initial angle, angular frequency, orbital eccentricity, and asphericity. Synchronisation between orbital and rotational dynamics, low eccentricity, and reduced asphericity stabilise the motion, while higher values amplify chaos. The deterministic but highly sensitive nature of Hyperion’s rotation highlights the strong coupling between its orbital and rotational dynamics, providing insights into the mechanisms behind its unique chaotic motion.

C. Poincaré Sections

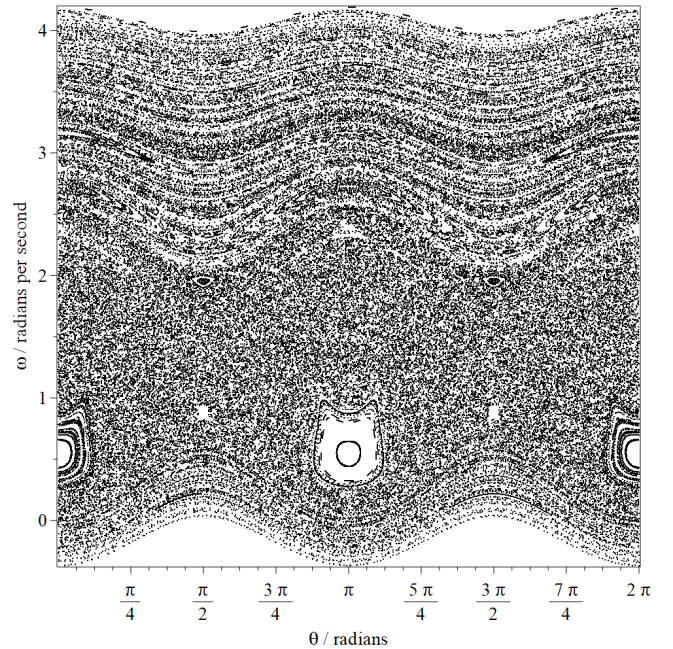


FIG. 8. Poincaré section for Hyperion with $e = 0.105$ and $\beta = 0.89$. $\omega(t)$ is a multiple of the average orbital frequency ω_H . 1000 initial conditions were taken and plotted for 100 orbits each [3].

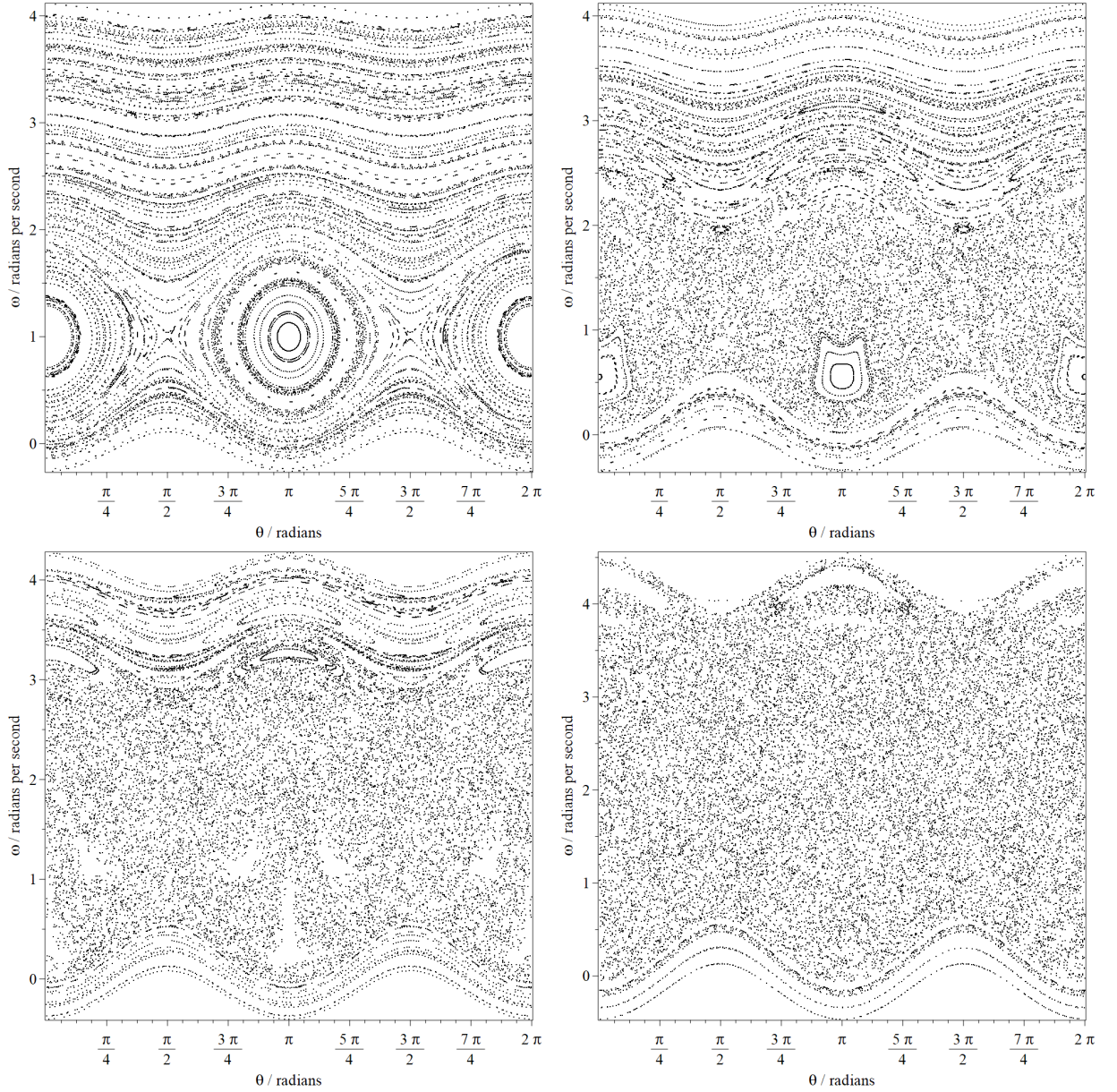


FIG. 9. Poincaré sections for varying eccentricities e with $\beta = 0.89$: **Top-Left:** $e = 0.0$, **Top-Right:** $e = 0.1$, **Bottom-Left:** $e = 0.2$, **Bottom-Right:** $e = 0.3$. Higher e introduces irregularities, transitioning to chaotic behaviour.

Poincaré sections are used to analyse Hyperion's rotational dynamics and identify regions of stability or chaos. Periodic orbits appear as a finite number of distinct points, indicating stable, repetitive motion, while quasi-periodic orbits form smooth, closed curves, representing bounded but non-repeating behaviour. In contrast, chaotic motion manifests as a scattered or dense cloud of points, reflecting the system's sensitivity to initial conditions and its aperiodic but deterministic nature. These sections plot the angular frequency ω against the orientation θ at specific orbital points, such as periapsis, revealing the phase space structure under varying parameters like orbital eccentricity e and asphericity β .

Figure 8 shows a Poincaré section for Hyperion's typical parameters, $e = 0.105$ and $\beta = 0.89$. The section shows near-stable regions every $\pi/2$ rads. It also shows that at higher frequencies it becomes more stable and the points align into waves, this is confirmed by Figure 5.

Figure 9 illustrates how orbital eccentricity e affects the phase space for a fixed $\beta = 0.89$. At $e = 0.0$ (top-left), corresponding to a circular orbit, the system shows regular behaviour with closed, ordered curves. As e increases to 0.1, 0.2, and 0.3, the phase space becomes increasingly irregular, culminating in significant scattering at $e = 0.3$, indicative of chaotic motion. This confirms that higher e destabilises rotational dynamics by increas-

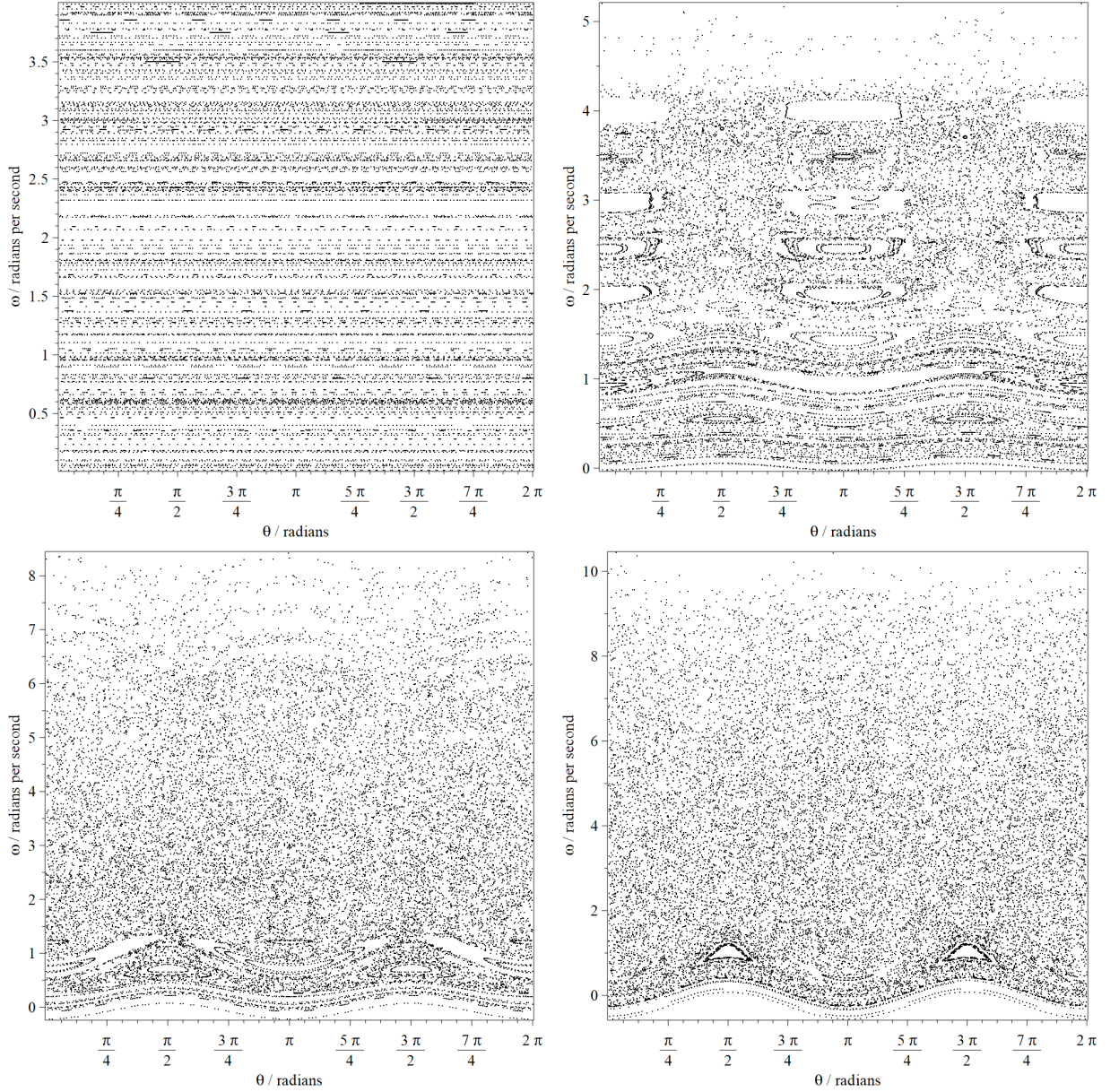


FIG. 10. Poincaré sections for varying asphericity β with $e = 0.105$: **Top-Left:** $\beta = 0.0$, **Top-Right:** $\beta = 0.2$, **Bottom-Left:** $\beta = 0.4$, **Bottom-Right:** $\beta = 0.6$. Higher β increases chaos and instability in the phase space.

ing torque variability.

Figure 10 examines the effect of asphericity β for $e = 0.105$. At $\beta = 0.0$ (top-left), representing a spherical body, the system exhibits periodic and stable behaviour. As β increases to 0.2, 0.4, and 0.6, the phase space becomes increasingly scattered, indicating growing instability. Greater asphericity amplifies gravitational torques, promoting chaos.

These results demonstrate that circular orbits and near-spherical shapes promote stable motion, while higher eccentricity and greater asphericity drive chaotic behaviour. This confirms that the interplay between e and β determines the degree of chaos in Hyperion's rota-

tional dynamics.

D. Fourier Analysis

The Fourier transform of $\omega(t)$ is used to analyse Hyperion's rotational dynamics and distinguish between periodic, quasi-periodic, and chaotic behaviour as system parameters, such as asphericity (β) and orbital eccentricity (e), vary.

Figure 11 presents the Fourier spectra for variations in β and e . The spectra help visualise how the dynamics transition from regular to chaotic as these parameters

increase.

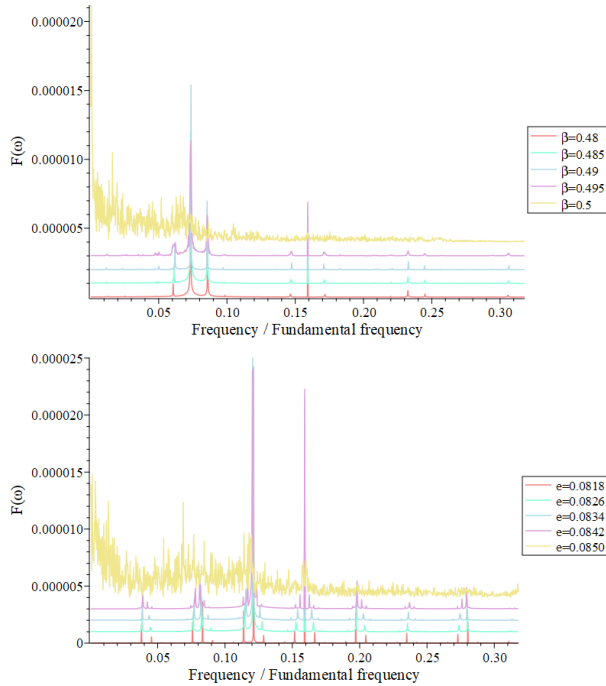


FIG. 11. **Top:** Fourier spectra of $\omega(t)$ for varying asphericity parameters β with fixed $e = 0.105$. Lower β values show sharp peaks, indicating regular motion, while higher β broadens the spectra and introduces noise, signalling chaotic dynamics. **Bottom:** Fourier spectra of $\omega(t)$ for varying orbital eccentricities e with fixed $\beta = 0.89$. Lower e values produce discrete peaks, indicating regular motion, while higher e broadens the spectra and adds frequencies, indicating chaotic behaviour.

Periodic motion corresponds to spectra with discrete peaks, reflecting stable, regular behaviour. Chaotic motion produces broader, noisier spectra with continuous frequency distributions, reflecting irregular dynamics. The top panel of Figure 11 shows that increasing β broadens the spectra and introduces additional frequencies, marking a transition to chaos. Similarly, the bottom panel demonstrates that increasing e destabilises the dynamics, leading to chaotic behaviour.

These spectra confirm that Hyperion’s chaotic rotation results from high asphericity and orbital eccentricity. The broadening of peaks with increasing β and e indicates a transition from regular, periodic motion to chaotic dynamics. This analysis supports theoretical predictions of chaos in non-linear systems, with period-doubling and aperiodic motion as key mechanisms. Period-doubling is evident in Figure 11: the bottom panel exhibits twice the periods of the top panel, with evenly spaced peaks. This behaviour arises due to the higher eccentricity and asphericity values in the bottom panel, which drive the system closer to chaos. Additionally, the bifurcation of peaks is visible, highlighting

the transition from regular motion to chaotic dynamics.

In summary, Fourier analysis provides a clear and efficient method to characterise transitions to chaos in Hyperion’s dynamics. The results demonstrate how increasing β and e destabilises the system, leading to chaotic behaviour.

IV. CONCLUSION

This study explored the chaotic rotational dynamics of Hyperion, focusing on the influence of initial angle (θ), angular frequency (ω), orbital eccentricity (e), and asphericity (β). Hyperion’s motion was found to be highly sensitive to initial conditions, with small changes in $\theta(0)$ or $\omega(0)$ resulting in significant divergence in rotational trajectories, a hallmark of chaos. However, specific configurations, such as angular frequencies near integer multiples of the orbital frequency (ω_H) or initial angles as multiples of $\pi/2$, demonstrated more stable behaviour due to synchronisation between rotational and orbital dynamics.

The effects of orbital eccentricity and asphericity were systematically examined. Higher e increased gravitational torque variability, destabilising the rotation and promoting chaos, whereas lower e produced more regular and stable motion. Similarly, higher β amplified torques, leading to chaotic dynamics, while lower β , representing a more spherical shape, stabilised the system.

Poincaré sections illustrated the transition from stability to chaos, showing periodic or quasi-periodic patterns at low e and β , and scattered, chaotic structures at higher values. Fourier analysis further revealed the mechanisms of chaos, with discrete peaks in the spectra indicating regular motion for low e and β , and broader spectra with additional frequencies reflecting chaotic behaviour as e and β increased. Evidence of period-doubling bifurcations confirmed theoretical predictions of chaos in non-linear systems.

These findings also provide a framework for understanding the rotational dynamics of other moons. For example, Phobos, with lower eccentricity ($e = 0.0151$) and asphericity ($\beta = 0.62$), is expected to exhibit stable and predictable motion. This analysis can be extended to explore transitions between regular and chaotic behaviour across various celestial systems.

In conclusion, Hyperion’s chaotic rotation is driven by its sensitivity to initial conditions, combined with high orbital eccentricity and asphericity. Numerical modelling, Poincaré sections, and Fourier analysis collectively provided valuable insights into the mechanisms governing Hyperion’s dynamics and those of other moons in the solar system.

V. REFERENCES

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- [1] R. L. Devaney, *An Introduction to Chaotic Dynamical Systems*, 2nd ed. (Westview Press, 2003).
- [2] C.-E. E. Meador, *Numerical Calculation of Lyapunov Exponents for Three-Dimensional Systems of Ordinary Differential Equations*, Ph.D. thesis, Marshall University (2011), theses, Dissertations and Capstones, Paper 107.
- [3] P. T. Boyd and S. J. Peale, The Astrophysical Journal **431**, 425 (1994).

Appendix

Investigation of Chaotic Dynamics in Rotation

> # Define Constants

$G := 6.6743 \cdot 10^{-11}$: # Gravitational constant (Nm^2/kg^2)

$MSat := 5.6832 \cdot 10^{26}$: # Mass of Saturn (kg)

$MH := 5.551 \cdot 10^{18}$: # Mass of Hyperion (kg)

$M := MSat + MH$: # Total mass (kg)

$a := 1.4815 \cdot 10^9$: # Semi-major axis (m)

$e := 0.105$: # Eccentricity

$\beta := 0.89$: # Asphersity

Orbital period

$$TH := \sqrt{\frac{4 \cdot \pi^2 \cdot a^3}{G \cdot M}} :$$

> $\text{plotsol} := \text{proc}(\text{var_num}, \text{theta0}, \text{omega0}, e, \beta, n, c)$

local $xH0, yH0, vxH0, vyH0, oH0, wH0, xH, yH, vxH, vyH, oH, wH, t, \text{Eqns}, \text{ICs}, \text{soln}, \text{variables}, i :$

global $G, MSat, MH, M, a, TH :$

Initial conditions

$xH0 := a \cdot (1 - e)$: # Initial x position (perihelion)

$yH0 := 0$: # Initial y position

$vxH0 := 0$: # Initial x velocity

$vyH0 := \sqrt{\frac{G \cdot M \cdot (1 + e)}{a \cdot (1 - e)}}$: # Initial y velocity

$oH0 := \text{theta0}$:

$wH0 := \text{omega0}$:

Define the new equations of motion

$\text{Eqns} := \text{diff}(xH(t), t) = vxH(t),$

$\text{diff}(yH(t), t) = vyH(t),$

$\text{diff}(vxH(t), t) = -\frac{G \cdot M \cdot xH(t)}{(xH(t)^2 + yH(t)^2)^{\frac{3}{2}}},$

$\text{diff}(vyH(t), t) = -\frac{G \cdot M \cdot yH(t)}{(xH(t)^2 + yH(t)^2)^{\frac{3}{2}}},$

$\text{diff}(oH(t), t) = wH(t),$

$\text{diff}(wH(t), t) =$

$-\frac{G \cdot MSat \cdot \beta^2 \cdot (xH(t) \cdot \sin(oH(t)) - yH(t) \cdot \cos(oH(t))) \cdot (xH(t) \cdot \cos(oH(t)) + yH(t) \cdot \sin(oH(t)))}{(xH(t)^2 + yH(t)^2)^{\frac{5}{2}}} :$

$\text{ICs} := xH(0) = xH0, yH(0) = yH0, vxH(0) = vxH0, vyH(0) = vyH0, oH(0) = oH0, wH(0) = wH0 :$

$\text{soln} := \text{dsolve}(\{\text{Eqns}, \text{ICs}\}, \text{numeric}, \text{method} = \text{rkf45}, \text{abserr} = 10^{-10}, \text{relerr} = 10^{-10}, \text{output} = \text{listprocedure}, \text{maxfun} = 0) :$

```
#soln := dsolve( {Eqns,ICs}, numeric, output=listprocedure) :
```

```
variables := [xH(t),yH(t),vxH(t),vyH(t),oH(t),wH(t)] :
```

```
odeplot(soln,[t,variables[var_num]],0..n*TH,numpoints=5000,tickmarks=[[seq(i*TH=i,i=0..n)],default],color=c);
```

```
end proc:
```

```
|
```

```
> cols := ["LightCoral","Aquamarine","LightBlue","Plum","Khaki","Orchid","PaleGreen"] :
```

```
p1 := plotsol(6,0,2*Pi/TH,e,beta,20,cols[1]) :
```

```
p2 := plotsol(6,0.01,2*Pi/TH,e,beta,20,cols[2]) :
```

```
p3 := plotsol(6,0.02,2*Pi/TH,e,beta,20,cols[3]) :
```

```
p4 := plotsol(6,0.03,2*Pi/TH,e,beta,20,cols[4]) :
```

```
display([p1,p2,p3,p4],labels=["Orbital Periods","ω(t) / radians per second"],labeldirections=[horizontal,vertical],
legend=["θ=0, ω=2π/TH","θ=0.01, ω=2π/TH","θ=0.02, ω=2π/TH","θ=0.03, ω=2π/TH"],legendstyle=[location=right,font=[default,13]],labelfont=[default,16],axesfont=[default,14],size=[800,450]);
```

Poincare Sections

```
> poincare := proc(theta0, omega0, e, beta, n)
```

```
local xH0,yH0,vxH0,vyH0,oH0,wH0,xH,yH,vxH,vyH,oH,wH,t,Eqns,ICs,soln,i :
```

```
global G,MSat,MH,M,a,TH :
```

```
# Initial conditions
```

```
xH0 := a*(1 - e) : # Initial x position (perihelion)
```

```
yH0 := 0 : # Initial y position
```

```
vxH0 := 0 : # Initial x velocity
```

```
vyH0 := sqrt(G*M*(1+e)/(a*(1-e))) : # Initial y velocity
```

```
oH0 := theta0 :
```

```
wH0 := omega0 :
```

```
# Define the new equations of motion
```

```
Eqns := diff(xH(t),t)=vxH(t),
```

```
diff(yH(t),t)=vyH(t),
```

$$\text{diff}(vxH(t),t) = -\frac{G \cdot M \cdot xH(t)}{(xH(t)^2 + yH(t)^2)^{\frac{3}{2}}},$$

$$\text{diff}(vyH(t),t) = -\frac{G \cdot M \cdot yH(t)}{(xH(t)^2 + yH(t)^2)^{\frac{3}{2}}},$$

$$\text{diff}(oH(t),t) = wH(t),$$

$$\text{diff}(wH(t),t) =$$

$$- \frac{G \cdot MSat \cdot \beta^2 \cdot (xH(t) \cdot \sin(oH(t)) - yH(t) \cdot \cos(oH(t))) \cdot (xH(t) \cdot \cos(oH(t)) + yH(t) \cdot \sin(oH(t)))}{(xH(t)^2 + yH(t)^2)^{\frac{5}{2}}} :$$

$ICs := xH(0) = xH0, yH(0) = yH0, vxH(0) = vxH0, vyH(0) = vyH0, oH(0) = oH0, wH(0) = wH0 :$

$soln := dsolve(\{Eqns, ICs\}, numeric, method = rkf45, abserr = 10^{-11}, relerr = 10^{-12}, output = listprocedure, maxfun = 0) :$

$\#soln := dsolve(\{Eqns, ICs\}, numeric, output = listprocedure, maxfun = 0) :$

$odeplot(soln, \left[evalf\left(oH(t) - 2 \cdot \text{Pi} \cdot \text{floor}\left(\frac{oH(t)}{2 \cdot \text{Pi}}\right)\right), \frac{wH(t) \cdot TH}{2 \cdot \text{Pi}} \right], 0 .. n \cdot TH, numpoints = n + 1, style = point, symbol = point, labels = ["\theta / \text{radians}", "\omega / \text{radians per second}"], labeldirections = [horizontal, vertical], color = black, tickmarks = [piticks, default], axes = boxed, size = [1000, 1000]) ;$

end proc:

> $plots_list := [] : \# \text{ Initialize an empty list for storing plots}$

for i **from** 1 **to** 1000 **do**

$theta0 := rand(0 .. evalf(2 \cdot \text{Pi}))() :$

$omega0 := rand\left(0 .. evalf\left(\frac{8 \cdot \text{Pi}}{TH}\right)\right)() :$

$plots_list := [op(plots_list), \text{poincare}(theta0, omega0, e, \beta, 100)] :$

end do:

$display(plots_list, axesfont = [default, 22], labelfont = [default, 22]) ;$

Fourier Analysis

> $fourier := \text{proc}(theta0, omega0, e, \beta, n, c, offset)$

local $xH0, yH0, vxH0, vyH0, oH0, wH0, xH, yH, vxH, vyH, oH, wH, t, Eqns, ICs, soln, variables, i, w_values, fft_result, frequencies, N :$

global $G, MSat, MH, M, a, TH :$

$\# \text{ Initial conditions}$

$xH0 := a \cdot (1 - e) : \# \text{ Initial } x \text{ position (perihelion)}$

$yH0 := 0 : \# \text{ Initial } y \text{ position}$

$vxH0 := 0 : \# \text{ Initial } x \text{ velocity}$

$vyH0 := \sqrt{\frac{G \cdot M \cdot (1 + e)}{a \cdot (1 - e)}} : \# \text{ Initial } y \text{ velocity}$

$oH0 := theta0 :$

$wH0 := omega0 :$

$\# \text{ Define the new equations of motion}$

$Eqns := \text{diff}(xH(t), t) = vxH(t),$

$\text{diff}(yH(t), t) = vyH(t),$

$$\begin{aligned}
diff(vxH(t), t) &= - \frac{G \cdot M \cdot xH(t)}{(xH(t)^2 + yH(t)^2)^{\frac{3}{2}}}, \\
diff(vyH(t), t) &= - \frac{G \cdot M \cdot yH(t)}{(xH(t)^2 + yH(t)^2)^{\frac{3}{2}}}, \\
diff(oH(t), t) &= wH(t), \\
diff(wH(t), t) &= \\
&- \frac{G \cdot MSat \cdot \beta^2 \cdot (xH(t) \cdot \sin(oH(t)) - yH(t) \cdot \cos(oH(t))) \cdot (xH(t) \cdot \cos(oH(t)) + yH(t) \cdot \sin(oH(t)))}{(xH(t)^2 + yH(t)^2)^{\frac{5}{2}}}.
\end{aligned}$$

$ICs := xH(0) = xH0, yH(0) = yH0, vxH(0) = vxH0, vyH(0) = vyH0, oH(0) = oH0, wH(0) = wH0 :$

$soln := dsolve(\{Eqns, ICs\}, numeric, method = rkf45, abserr = 10^{-10}, relerr = 10^{-10}, output = listprocedure, maxfun = 0) :$

$N := 2^{11} : \# \text{ Number of samples}$

$w_values := \text{Array}(1..N, datatype = float[8]) :$

for i **from** 1 **to** N **do**

$t := \frac{(i - 1) \cdot n \cdot TH}{N} :$

$w_values[i] := evalf(rhs(soln[5](t))) :$

end do:

$fft_result := FFT(w_values) :$

$frequencies := \left[seq\left(\frac{i}{n \cdot TH}, i = 0 .. \frac{N}{2} - 1\right) \right] :$

$plot\left(\left[seq\left(\left[\frac{frequencies[i] \cdot TH}{2 \cdot \Pi}, abs(fft_result[i]) + offset \right], i = 2 .. \frac{N}{2} \right) \right], color = c, style = line, labels = ["Frequency / Fundamental frequency", "F(\omega)"], labeldirections = [horizontal, vertical] \right);$

end proc:

> $p1 := \text{fourier}(0, 1.025 * (2 * \Pi / TH), 0.105, 0.48, 512, cols[1], 0) :$
 $p2 := \text{fourier}(0, 1.025 * (2 * \Pi / TH), 0.105, 0.485, 512, cols[2], 0.1e-5) :$
 $p3 := \text{fourier}(0, 1.025 * (2 * \Pi / TH), 0.105, 0.49, 512, cols[3], 0.2e-5) :$
 $p4 := \text{fourier}(0, 1.025 * (2 * \Pi / TH), 0.105, 0.495, 512, cols[4], 0.3e-5) :$
 $p5 := \text{fourier}(0, 1.025 * (2 * \Pi / TH), 0.105, 0.5, 512, cols[5], 0.4e-5) :$
 $display([p1, p2, p3, p4, p5], legend = ["\beta=0.48", "\beta=0.485", "\beta=0.49", "\beta=0.495", "\beta=0.5"], legendstyle = [location = right, font = [default, 13]], labelfont = [default, 16], axesfont = [default, 14], size = [800, 450]);$